

Supplementary Information

MVCL-DTI: Predicting Drug-Target Interactions Using a Multiview Contrastive Learning Model on a Heterogeneous Graph

Bei Zhang,^{†,‡} Lijun Quan,^{*,†,§} Zhijun Zhang,[†] Lexin Cao,[†] Qiufeng Chen,[†] Liangchen Peng,[†] Junkai

Wang,[†] Yelu Jiang,[†] Liangpeng Nie,[†] Geng Li,[†] Tingfang Wu^{†,§} and Qiang Lyu^{†,§}

[†]School of Computer Science and Technology, Soochow University, Jiangsu 215006, China

[‡]China Mobile (Suzhou) Software Technology Company Limited, Suzhou 215163, China

[§]Collaborative Innovation Center of Novel Software Technology and Industrialization, Jiangsu

210000, China

*E-mail: ljquan@suda.edu.cn

This supplementary file includes:

Figure S1 AUPR of different ratios of positive and negative samples

Figure S2 Visualization of drug features

Figure S3 Visualization of node-specific attention weights

Figure S4 Visualization of view-specific attention weights

Figure S5 The ROC curve of MVCL-DTI with different layers and dimensions.

Table S1 AUC, AUPR, accuracy, recall, fl, mcc of benchmarks with a 1:1 positive-to-negative sample ratio

Table S2 AUC, AUPR, accuracy, recall, fl, mcc of benchmarks with a 1:10 positive-to-negative sample ratio

Table S3 Compared with other methods, when the proportion of positive and negative samples increases, the decrease in AUPR of MVCL-DTI is relatively small.

Table S4 Pairwise Statistical Comparisons of AUROC Scores Using the Nemenyi Test with a 1:1/1:10 positive-to-negative sample ratio

Table S5 Pairwise Statistical Comparisons of AUPR Scores Using the Nemenyi Test with a 1:1/1:10 positive-to-negative sample ratio

Table S6 AUROC Scores with Mean, Standard Deviation, and 95% Confidence Intervals with a 1:1/1:10 positive-to-negative sample ratio

Table S7 AUPR Scores with Mean, Standard Deviation, and 95% Confidence Intervals with a 1:1/1:10 positive-to-negative sample ratio

Table S8 The targets we collected from DrugBank and pubchem that interact with four COVID-19 drugs and are present in the D1936 dataset, as well as the interaction scores predicted by MVCL-DTI.

Table S9 Hyperparameters settings for MVCL-DTI

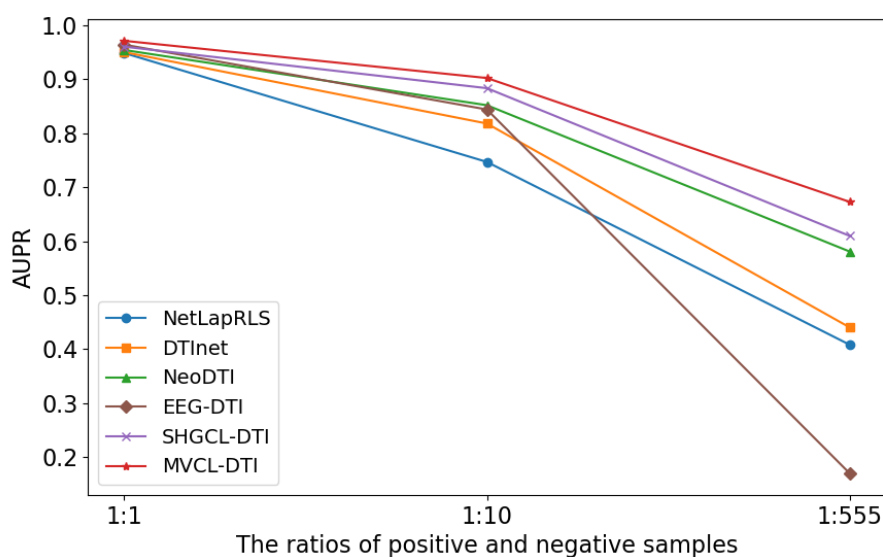


Figure S1. AUPR of different ratios of positive and negative samples. AUPR gradually decreases as the ratio of negative samples increases.

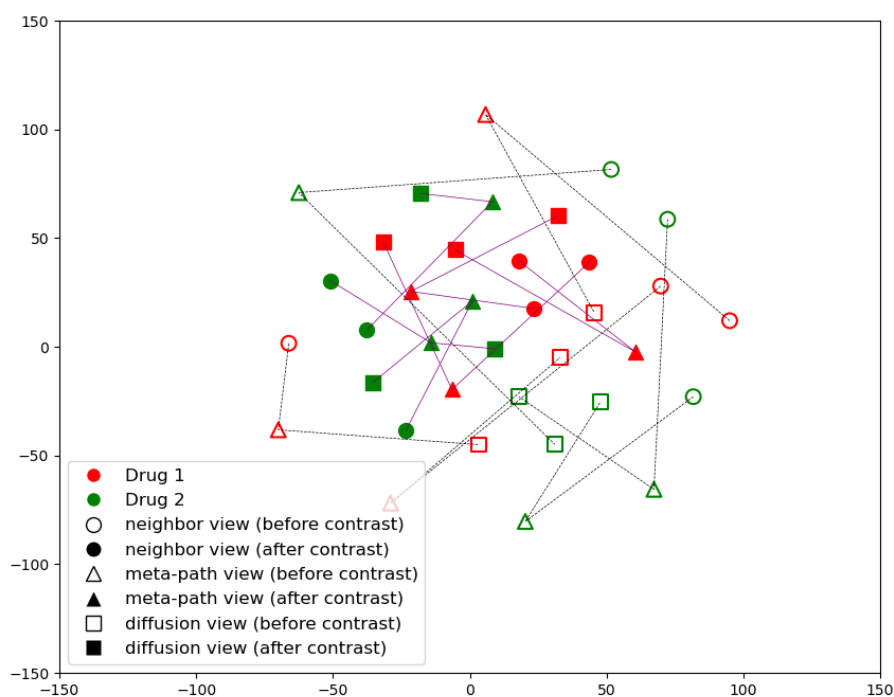


Figure S2. Visualization of drug features. Red represents drug1, which includes Methylprednisolone, Fluoxymesterone, Spironolactone; green represents drug2, which includes Triazolam, Midazolam, Alprazolam. Hollow and solid circles indicate the neighbor view before and after contrastive learning, respectively; hollow and solid triangles represent the meta-path view

before and after contrastive learning; hollow and solid squares correspond to the diffusion view before and after contrastive learning. Black dashed lines and purple solid lines connect the same protein across the three views, before and after contrastive learning, respectively.

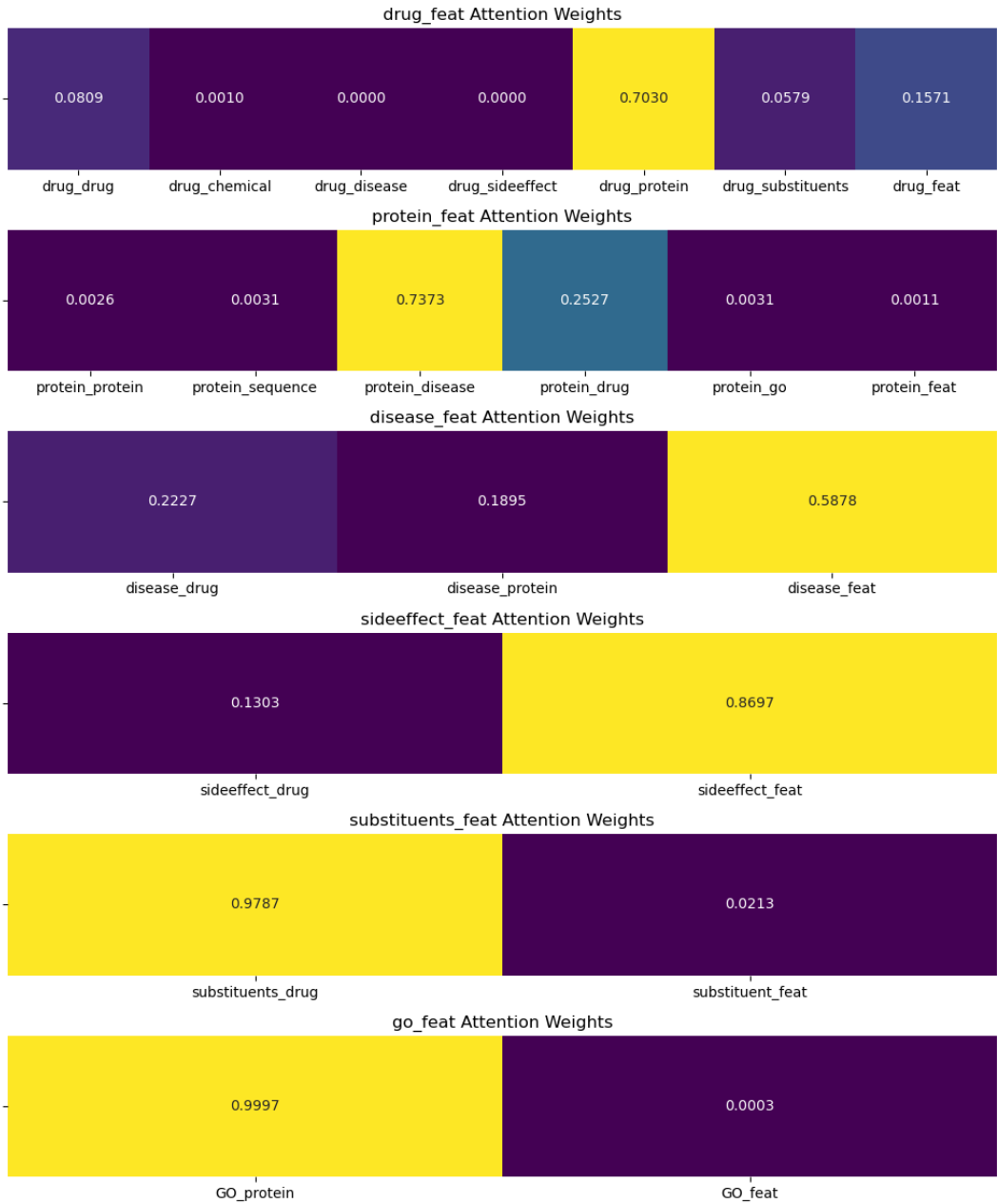


Figure S3. Visualization of node-specific attention weights. This figure illustrates the attention weights learned during feature fusion in the diffusion view, offering insights into how the model integrates information from different node types and relationships. The heatmaps depict the contribution of specific interactions to the final feature representations.

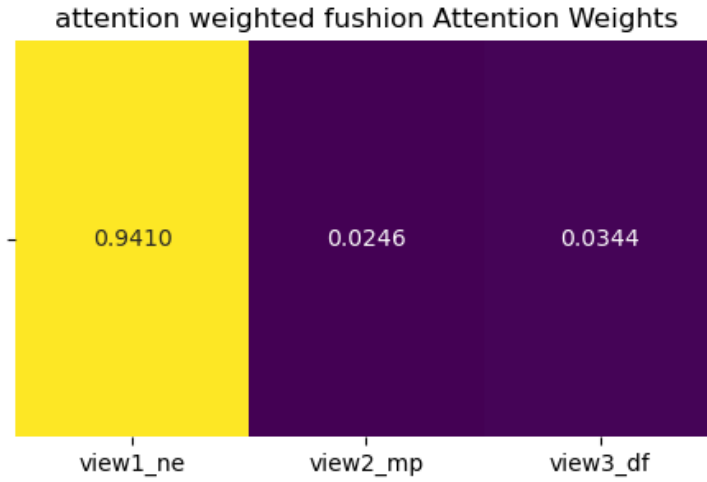


Figure S4. Visualization of view-specific attention weights. This figure showcases the attention weights assigned to the three views—neighbor, meta-path, and diffusion—during the multi-view attention weighted fusion process.

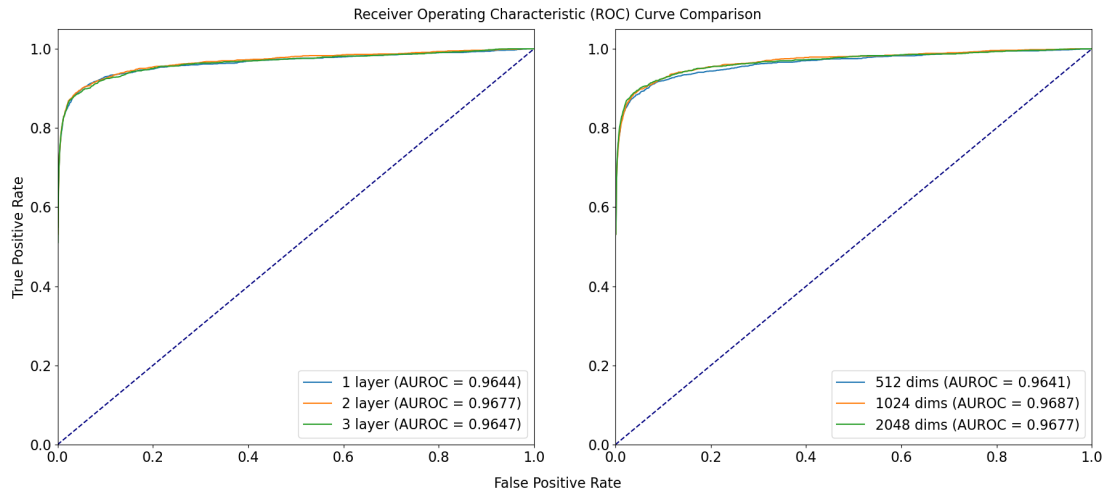


Figure S5. The ROC curve of MVCL-DTI with different layers and dimensions. A sensitivity analysis of the model's hyperparameters is conducted to identify the optimal structure, focusing on: (a) determining the ideal number of layers for each view encoder, and (b) experimenting with different hidden layer dimensions within the network.

Table S1. AUC, AUPR, accuracy, recall, fl, mcc of benchmarks with a 1:1 positive-to-negative

	sample ratio						
	AUC	AUPR	accuracy	precision	recall	fl	mcc
NetLapRLS	0.9412	0.9485	0.8677	0.8881	0.8628	0.8754	0.7264
DTInet	0.9302	0.9538	0.8557	0.8710	0.8876	0.8793	0.7326
NeoDTI	0.9455	0.9555	0.8781	0.8953	0.8579	0.8762	0.7583
EEG-DTI	0.9551	0.9643	0.9009	0.584	0.9891	0.7335	0.8061
SHGCL-DTI	0.9510	0.9621	0.9012	0.9228	0.8715	0.8984	0.8049
MVCL-DTI	0.9648	0.9714	0.9171	0.9236	0.9048	0.9164	0.8352

Table S2. AUC, AUPR, accuracy, recall, fl, mcc of benchmarks with a 1:10 positive-to-negative

	sample ratio						
	AUC	AUPR	accuracy	precision	recall	fl	mcc
NetLapRLS	0.9073	0.7406	0.8570	0.8485	0.4404	0.5798	0.5977
DTInet	0.9150	0.8184	0.9116	0.8822	0.7980	0.8380	0.7031
NeoDTI	0.9452	0.8495	0.9569	0.8819	0.7357	0.8023	0.7882
EEG-DTI	0.9554	0.8747	0.8023	0.6741	0.8906	0.7262	0.8082
SHGCL-DTI	0.9582	0.8851	0.9724	0.8981	0.7821	0.8384	0.8261
MVCL-DTI	0.9677	0.9024	0.9749	0.8992	0.8107	0.8549	0.8429

Table S3. Compared with other methods, when the proportion of positive and negative samples

increases, the decrease in AUPR of MVCL-DTI is relatively small.

Ratio	1: 1	1: 10	1: all
	AUPR	AUPR	AUPR
NetLapRLS	0.9485	0.7406 ↓ 0.2079	0.4081 ↓ 0.3325
DTInet	0.9538	0.8184 ↓ 0.1354	0.4402 ↓ 0.3782
NeoDTI	0.9555	0.8495 ↓ 0.106	0.5808 ↓ 0.2687
EEG-DTI	0.9643	0.8747 ↓ 0.0896	0.1697 ↓ 0.705
SHGCL-DTI	0.9621	0.8851 ↓ 0.077	0.6098 ↓ 0.2738
MVCL-DTI	0.9714	0.9024 ↓ 0.069	0.6728 ↓ 0.2296

Table S4. Pairwise Statistical Comparisons of AUROC Scores Using the Nemenyi Test with a
1:1/1:10 positive-to-negative sample ratio

Comparison	p-value	Adjusted p-value	Significant (p < 0.05)
NetLapRLS	0.002/0.002	0.029/0.029	Yes/Yes
DTInet	0.006/0.002	0.041/0.029	Yes/Yes
NeoDTI	0.002/0.002	0.029/0.029	Yes/Yes
EEG-DTI	0.010/0.010	0.039/0.049	Yes/Yes
SHGCL-DTI	0.002/0.006	0.029/0.035	Yes/Yes

Table S5. Pairwise Statistical Comparisons of AUPR Scores Using the Nemenyi Test with a
1:1/1:10 positive-to-negative sample ratio

Comparison	p-value	Adjusted p-value	Significant (p < 0.05)
NetLapRLS	0.002/0.002	0.029/0.029	Yes/Yes
DTInet	0.002/0.002	0.029/0.029	Yes/Yes
NeoDTI	0.002/0.002	0.029/0.029	Yes/Yes
EEG-DTI	0.002/0.002	0.029/0.029	Yes/Yes
SHGCL-DTI	0.004/0.010	0.029/0.029	Yes/Yes

Table S6. AUROC Scores with Mean, Standard Deviation, and 95% Confidence Intervals with a
1:1/1:10 positive-to-negative sample ratio

Methods	Mean $\pm$ Std	95% Confidence Interval
NetLapRLS	0.9412 $\pm$ 0.010/0.9073 $\pm$ 0.015	[0.935, 0.944]/ [0.896,0.917]
DTInet	0.9302 $\pm$ 0.009/0.9150 $\pm$ 0.017	[0.923, 0.935]/ [0.903,0.927]
NeoDTI	0.9455 $\pm$ 0.008/0.9452 $\pm$ 0.011	[0.939, 0.950]/ [0.937,0.953]
EEG-DTI	0.9551 $\pm$ 0.011/0.9554 $\pm$ 0.012	[0.947, 0.963]/ [0.949,0.962]
SHGCL-DTI	0.9510 $\pm$ 0.013/0.9582 $\pm$ 0.011	[0.941, 0.961]/ [0.951,0.966]
MVCL-DTI	0.9648$\pm$ 0.008/0.9677$\pm$ 0.011	[0.959, 0.971]/ [0.960,0.976]

Table S7. AUPR Scores with Mean, Standard Deviation, and 95% Confidence Intervals with a
1:1/1:10 positive-to-negative sample ratio

Methods	Mean $\pm$ Std	95% Confidence Interval
NetLapRLS	0.9485 $\pm$ 0.010/0.7406 $\pm$ 0.022	[0.935, 0.951]/[0.726,0.757]
DTInet	0.9538 $\pm$ 0.011/0.8184 $\pm$ 0.016	[0.941, 0.958]/[0.806,0.829]
NeoDTI	0.9555 $\pm$ 0.008/0.8495 $\pm$ 0.014	[0.944, 0.956]/[0.840,0.860]
EEG-DTI	0.9643 $\pm$ 0.011/0.8747 $\pm$ 0.012	[0.949, 0.965]/[0.869,0.878]
SHGCL-DTI	0.9621 $\pm$ 0.010/0.8851 $\pm$ 0.020	[0.955, 0.969]/[0.871,0.900]
MVCL-DTI	0.9714$\pm$ 0.007/0.9024$\pm$ 0.010	[0.967, 0.976]/ [0.888,0.917]

Table S8. The targets we collected from DrugBank and pubchem that interact with four COVID-19 drugs and are present in the D1936 dataset, as well as the interaction scores predicted by

MVCL-DTI.

Drug	Drugbank id	Prediction score	Target	Existence basis	Whether the interaction exists in the D1936
Dexamethasone	DB01234	0.916	NR3C1	DrugBank	Yes
		0.8204	NR0B1	DrugBank	Yes
		0.6317	ANXA1	DrugBank	Yes
		0.743	AR	pubchem	No
		0.7361	NR3C2	pubchem	No
		0.7098	ESR1	pubchem	No
		0.0847	NOS2	DrugBank	Yes
		0.3819	NR1I2	DrugBank	No
		0.0789	CYP3A4	DrugBank	No
		0.4865	CYP17A1	DrugBank	No
		0.2244	CYP11B1	DrugBank	No
		0.0395	ABCB1	DrugBank	No
		0.0012	ABCB11	DrugBank	No
		0	ABCC2	DrugBank	No
Methylprednisolone	DB00959	0.9005	NR3C1	DrugBank	Yes
		0.5699	ANXA1	DrugBank	No
		0.788	AR	pubchem	No
		0.656	ESR1	pubchem	No
		0.0208	ABCB1	DrugBank	No
		0	AKR1C1	DrugBank	No
		0.0029	AKR1C3	DrugBank	No
		0.0407	CYP3A4	DrugBank	No
		0.0107	CYP2C9	DrugBank	No
Ritonavir	DB00503	0.0962	NR1I2	DrugBank	No
		0	CYP2C9	DrugBank	No
		0	ABCB1	DrugBank	No
		0	ABCB11	DrugBank	No
		0	CYP3A4	DrugBank	No
		0	CYP1A2	DrugBank	No
		0	ABCC2	DrugBank	No
Darunavir	DB01264	0	CYP3A4	DrugBank	No
		0	ABCB1	DrugBank	No

Table S9. Hyperparameters settings for MVCL-DTI

Hyperparameters	Search space	Best parameter
Learning rate	[0.1, 0.01, 0.001]	0.001
Layers of three view encoders	[1, 2, 3]	2
Dimensions of hidden layers of network	[512, 1024, 2048]	2048
Temperature parameter	[0.25, 0.5, 0.75, 1]	0.5
Dropout rate	[0.1, 0.2, 0.3]	0.2
Number of iterations in propagation	[2, 3, 4]	3
Restart probability	[0.5, 0.7, 0.9]	0.9
Edge dropout in propagation	[0.3, 0.5, 0.7]	0.5